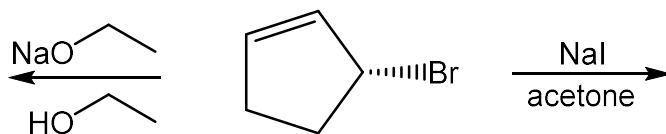


Nucleophilic Substitution and β -Elimination Worksheet Key

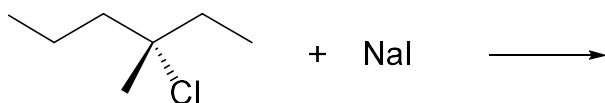
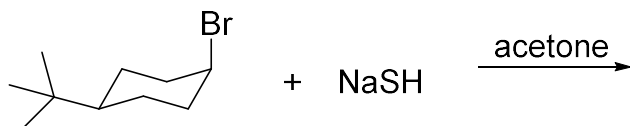
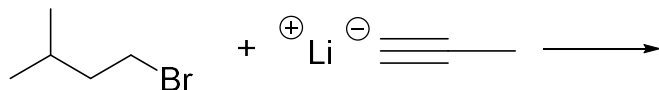
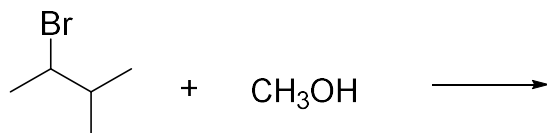
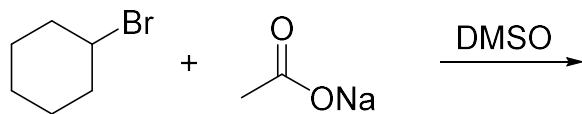
1. Draw a mechanism for the following reactions:



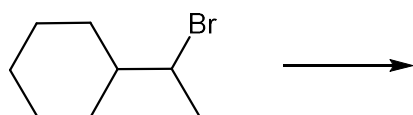
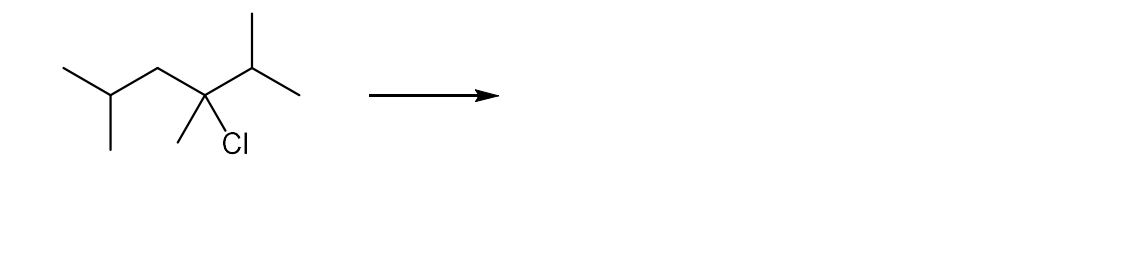
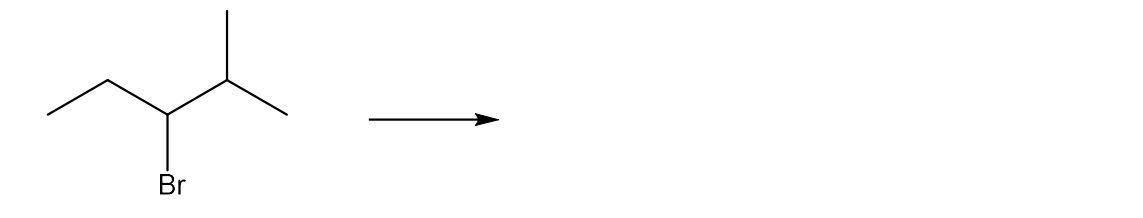
2. What product results from reaction of (S)-2-bromohexane and sodium acetate (NaOOCCH_3)? Draw the **reactant** and the **product** and indicate stereochemistry of **both**.

3. What *substitution* product would result from the solvolysis of (S)-3-bromo-3-methyloctane in methanol? **Draw** the **reactant** and the **product** and indicate stereochemistry of **both**.

4. Draw the major organic product of the following reactions and identify whether the reaction proceeded through an S_N1 or an S_N2 mechanism based on the degree of substitution of the substrate and the strength of the nucleophile.

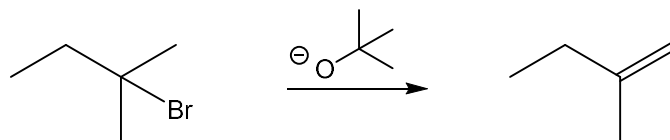


5. Draw **ALL** elimination products expected from the following alkyl halides. Circle those which can only be generated from an E1 mechanism.

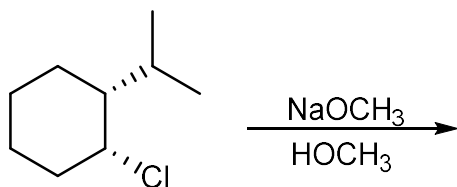
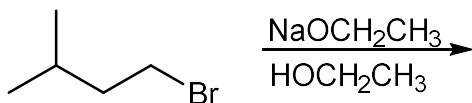
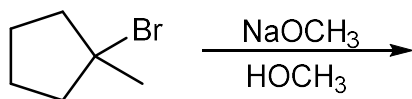
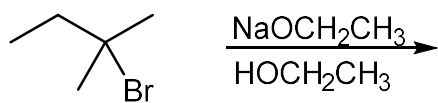


6. Sighting down C2 and C3 of (S)-2-bromobutane, draw a Newman projection that shows the proper orientation required for elimination to (Z)-but-2-ene.

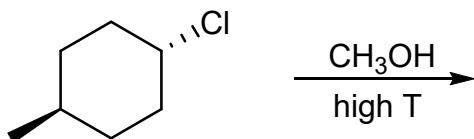
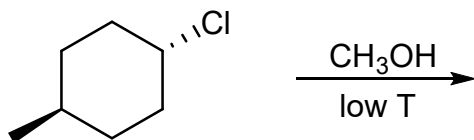
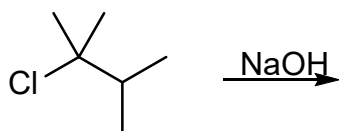
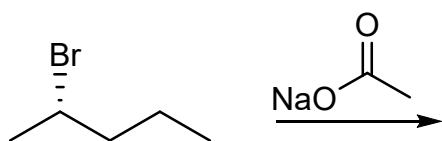
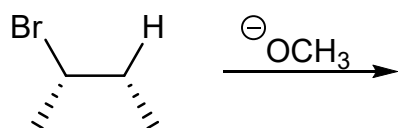
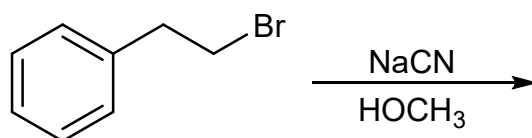
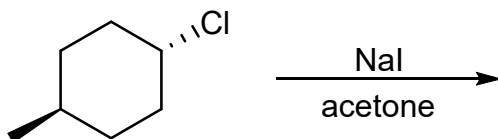
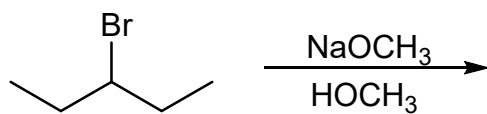
7. Draw a mechanism for the following reaction.



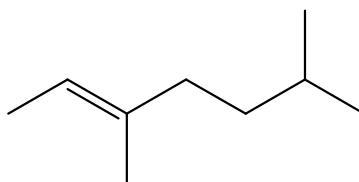
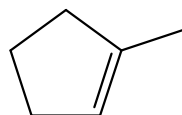
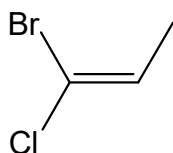
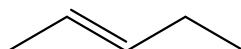
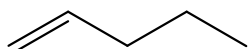
8. Draw the elimination products for the following and if a mixture of alkene products is predicted, circle the preferred product.



9. Draw the major organic product and predict whether each reaction proceeds by S_N1 , $E1$, S_N2 , or $E2$.



10. Write the IUPAC name for the following molecules.



11. Draw the line structure that corresponds to the following IUPAC names.

7-bromocyclohepta-1,3,5-triene

(E)-1-bromo-1-chloroprop-1-ene

3-bromo-(3Z,5E)-octa-3,5-diene

2-ethylcyclohexa-1,3-diene